If V is a vector space, a nonempty subset $U \subseteq V$ is called a subspace of V if U is itself a vector space using addition and scalar multiplication

$\vec{v} + \vec{w} \in U$ $\quad c\vec{v} \in U$ $\quad \vec{0} \in U$, where $\vec{0}$ is the zero vector in V

V is a subspace of itself

$U = \{\vec{0}\}$ is the zero subspace of V

If A is an m by n matrix A, the column space of A is subspace of $\mathbb{R}^m$. It is spanned by the columns of A

Example. Let $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ 4 & 3 \end{bmatrix}$ $C(A)$ is a subspace of $\mathbb{R}^3$ It is a plane containing the 2 columns of A.

$A\vec{x} = \vec{b}$ is solvable when $\vec{b} \in C(A)$, or $\vec{b}$ lies on this plane

$C(A)$ passes through $(0,0,0)$

There is certainly a solution $\vec{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ to $A\vec{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Eg. Given $\vec{b} = \begin{bmatrix} 1 \\ 8 \\ 10 \end{bmatrix} = -1\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + 2\begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix}$, then $A\vec{x} = \vec{b}$ admit a solution $\vec{x} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

Most $\vec{b} \in \mathbb{R}^3$ are not in $C(A)$. For those $\vec{b}$, there is no solution to $A\vec{x} = \vec{b}$.

Let S be a set of vectors in the vector space V.

Let SS be the set containing all the linear combinations of the vectors in S.

Then SS is a subspace of V spanned by S, or S spans SS.

When S is the set of columns of A, SS is $C(A)$.

SS is the smallest subspace of V containing S

4.2 The nullspace of A

Goal: when $\vec{b} = \vec{0}$, find the subspace of $\mathbb{R}^n$ containing all solutions to $A\vec{x} = \vec{0}$.

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For other matrices, there may be other nonzero sols.

$N(A)$ is a subspace of $\mathbb{R}^n$

Proof: For any matrix A, N

Suppose $\vec{x}, \vec{y} \in N(A)$

$A(\vec{x} + \vec{y}) = A$

So $\vec{x} + \vec{y}$, $c\vec{x} \in N(A)$

Example Describe the nullspa

Sol Apply elimination to

It yields $x_2 = 0$, and

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4.2 The nullspace of A

Goal: when $\vec{b} = \vec{0}$, find the subspace of $\mathbb{R}^n$ containing all solutions to $A\vec{x} = \vec{0}$.

For invertible matrices, $\vec{x} = \vec{0} \in \mathbb{R}^n$ is the only solution to $A\vec{x} = \vec{0}$. For other matrices, there may be other nonzero sols.

Definition The nullspace $N(A)$ of A consists of all solutions to $A\vec{x} = \vec{0}$. $\vec{0} \in N(A) \subset \mathbb{R}^n$

Q: How can we find the complete solution to $A\vec{x} = \vec{0}$? Elimination

Q: How to find the special solutions?

- The free components of $\vec{x}$ correspond the columns with NO pivots, i.e., the free columns.

$N(A)$ is a subspace of $\mathbb{R}^n$

Proof: For any matrix A, $N(A)$ is a nonempty subset in $\mathbb{R}^n$.

Suppose $\vec{x}, \vec{y} \in N(A)$. It means $A\vec{x} = \vec{0}$ and $A\vec{y} = \vec{0}$

$A(\vec{x}+\vec{y}) = A\vec{x} + A\vec{y} = \vec{0} + \vec{0} = \vec{0}$. $A(c\vec{x}) = c(A\vec{x}) = c\vec{0} = \vec{0}$

So $\vec{x}+\vec{y}, c\vec{x} \in N(A)$. It implies $N(A)$ is a subspace of $\mathbb{R}^n$

Example Describe the nullspace of $A_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $\quad U = \begin{bmatrix} \textcircled{1} & 2 \\ & \textcircled{-2} \end{bmatrix}$

Sol Apply elimination to $A_1\vec{x} = \vec{0}$ $\quad x_1 + 2x_2 = 0$, $3x_1 + 4x_2 = 0$ $\xrightarrow{r_2 - 3r_1}$ $x_1 + 2x_2 = 0$, $-2x_2 = 0$

It yields $x_2 = 0$, and then $x_1 = 0$ So $N(A_1) = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$

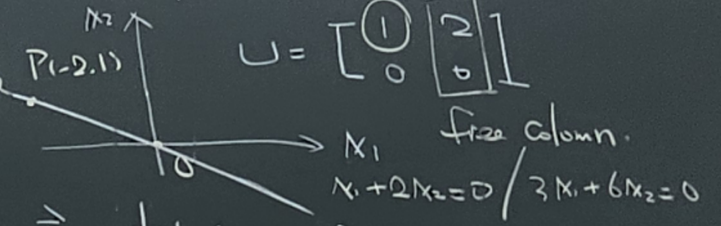
Example Describe the nullspace of $B = \begin{bmatrix} A_1 \\ -2A_1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 2 & 4 \\ 6 & 8 \end{bmatrix}$ $\quad B\vec{x} = \vec{0}$ $\quad N(B) = N(A_1) = \{\vec{0}\}$

For some matrices, the only solution to $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$ $\quad N(A) = \{\vec{0}\}$

This is caused by the fact that all columns of A have pivots, and no columns are "free".

Example. Describe the nullspace of $A_2 = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ $\quad U = \begin{bmatrix} \textcircled{1} & \boxed{2} \\ 0 & \boxed{0} \end{bmatrix}$

By elimination $\begin{matrix} x_1 + 2x_2 = 0 \\ 3x_1 + 6x_2 = 0 \end{matrix} \xrightarrow{r_2 - 3r_1} \begin{matrix} x_1 + 2x_2 = 0 \\ 0 = 0 \end{matrix}$



By specially choosing (free variable) $x_2 = 1$, x_1 is determined by $A_2\vec{x} = \vec{0}$, or $U\vec{x} = \vec{0}$, that $x_1 = -2$.

So $A_2\vec{x} = \vec{0}$ admits a special solution $\vec{s} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ $\quad N(A_2)$ consists of all multiples of $\vec{s}$.

When there are nonzero solutions to $A\vec{x} = \vec{0}$, $N(A)$ consists of all combinations of the special solutions to $A\vec{x} = \vec{0}$.

Q: How to find the special solutions?

The free components of $\vec{x}$ correspond the columns with NO pivots, i.e., the free columns.

$N(A)$ is a subspace of $\mathbb{R}^n$

Proof: For any matrix A, $N(A)$ is a nonempty subset in $\mathbb{R}^n$.

Suppose $\vec{x}, \vec{y} \in N(A)$, it means $A\vec{x} = \vec{0}$ and $A\vec{y} = \vec{0}$

$A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y} = \vec{0} + \vec{0} = \vec{0}$, $\quad A(c\vec{x}) = c(A\vec{x}) = c\vec{0} = \vec{0}$

Example $x+2y+3z=0$ $\quad A=[1\ 2\ 3]$ is of size 1 by 3.

The free variables are y and z

We can specially choose them as $y=1,\ z=0$ Then x can be uniquely determined by the eq as $x=-2$

$y=0,\ z=1$ $x=-3$

To obtain a special solution to $A\vec{x}=\vec{0}$, the special choices, 1 or 0, is only for the free variables

If the j-th column of A is free, there is a special solution to $A\vec{x}=\vec{0}$ with $x_j=1$, and other free variables being zeros.

$A\vec{x}=\begin{bmatrix}1&2&3\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=0$ has $\underbrace{2}_{3-1}$ special solutions $\vec{s}_1=\begin{bmatrix}-2\\1\\0\end{bmatrix}$ and $\vec{s}_2=\begin{bmatrix}-3\\0\\1\end{bmatrix}$

$A\vec{x}=\vec{0}$ produces a plane through the origin.

$N(A)$ is a plane. All vectors on the plane are combinations of $\vec{s}_1$ and $\vec{s}_2$

perpendicular to $\vec{n}=(1,2,3)$.

Example Describe the nullspace of $A=[A_1\ 2A_1]=\begin{bmatrix}1&2&2&4\\3&4&6&8\end{bmatrix}$ $\xrightarrow{r_2-3r_1}$ $\begin{bmatrix}\textcircled{1}&2&2&4\\0&\textcircled{-2}&0&-4\end{bmatrix}$

The free variables are x_3 and x_4 $\quad A\vec{x}=\vec{0}\ \rightarrow\ U\vec{x}=\vec{0}$

There are ... special solutions $\vec{s}_1=\begin{bmatrix}0\\1\\0\end{bmatrix}$ and $\vec{s}_2=\begin{bmatrix}\\0\\1\end{bmatrix}$

U free columns

4.2 The nullspace

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$A\vec{x} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$ has 2 special solutions $\vec{s}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ and $\vec{s}_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$ (3−1 = 2)

$A\vec{x} = \vec{0}$ produces a plane through the origin.

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Example Describe the nullspace of $A = [A_1 \; 2A_1] = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 3 & 4 & 6 & 8 \end{bmatrix}$ $\xrightarrow{r_2 - 3r_1}$ $\begin{bmatrix} \textcircled{1} & 2 & 2 & 4 \\ 0 & \textcircled{-2} & 0 & -4 \end{bmatrix}$ = U (free columns)

The free variables are x_3 and x_4. $A\vec{x} = \vec{0} \rightarrow U\vec{x} = \vec{0}$

There are $4 - 2 = 2$ special solutions (4: number of columns of A; 2: number of pivots) $\vec{s}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ and $\vec{s}_2 = \begin{bmatrix} 0 \\ -2 \\ 0 \\ 1 \end{bmatrix}$. So $N(A) = \ldots$

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